Comparison of Relational DBs to Graph Algorithms

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April 26, 2026

1 Introduction

2 Preliminaries

Let $\mathbb{C}$ be a universe of attribute names and VAL be a value domain, both infinite and mutually disjoint from each other and from any other set of symbols.

Definition 2.1 (Relational Database)**.** A *Relation Schema* R is a finite set of attribute names, i.e., $R \subseteq \mathbb{C}$. A *Database Schema* $\mathcal{S}$ is a finite set of relation schemas. A *Relation Instance* r over a relation schema R is a finite set of R*-rows*, where an R-row is a function $t : R \to \mathsf{VAL}$. A *Database Instance* RDB over a database schema $\mathcal{S}$ is a finite set of relation instances, one for each relation schema in $\mathcal{S}$. We denote the set of all RDBs by $\mathcal{RDB}$. We denote the set of all rows in RDB by $\mathsf{Dom}(\mathsf{RDB})$. We say that two rows t and t' are *joinable* if there is an attribute name A such that $t(A) = t'(A)$.

Definition 2.2 (Graph)**.** A *d-labeled undirected graph* is a tuple $G = (V, E, \rho)$ where V is a set of vertices and $E \subseteq \binom{V}{2}$ is a set of edges, both finite. If $(u, v) \in E$, we say that u and v are *adjacent* and we denote the *neighborhood* of u as $N(u) = \{v : \{u, v\} \in E\}$. ρ assigns a label to each vertex. Forammly, $\rho : V \to \mathbb{R}^d$. We denote the set of all d-labeled undirected graphs by $\mathcal{G}$.

In the theoretical part of the paper, in order to isolate the affect of the "structural information", we will focus on graphs with uniform labels.
The next definitions are from [**cucumides2026grablestabularlearningindependent**]

Definition 2.3 (row predictor)**.** A *row predictor* is a function rpred that maps each row in the RDB to a binary classification. $\mathsf{rpred}_{\mathsf{RDB}} : \mathsf{Dom}(\mathsf{RDB}) \to \{0, 1\}$.

Definition 2.4 (node predictor)**.** A *node predictor* is a function npred that maps each node in the graph G to a binary classification. $\mathsf{npred}_G : V \to \{0, 1\}$.

Definition 2.5 ($\mathsf{R2G}$ algorithm)**.** A *R2G algorithm* $\mathcal{A} : \mathcal{RDB} \to \mathcal{G}$ is a function that maps each RDB to a corresponding graph. Crucially, $\mathcal{A}(\mathsf{RDB})$ contains one distinct vertex for each row in RDB. We assume that every row has a corresponding vertex in the graph, i.e $V_t = \{v_t : t \in \mathsf{Dom}(\mathsf{RDB})\}$ and $V_t \subseteq V(\mathcal{A}(\mathsf{RDB}))$.

Definition 2.6 ($\mathsf{R2G}$ expressibility)**.** A row predictor rpred is $(\Gamma, \mathcal{P})$*-expressible* if there exist a $\mathsf{R2G}$ algorithm $\mathcal{A} \in \Gamma$ and a node predictor $\mathsf{npred} \in \mathcal{P}$ such that for every RDB, and for every $t \in \mathsf{Dom}(\mathsf{RDB})$, the following holds:

$$\mathsf{rpred}_{\mathsf{RDB}}(t) = \mathsf{npred}_{\mathcal{A}(\mathsf{RDB})}(v_t)$$

We write $\mathsf{Expr}(\Gamma, \mathcal{P}) = \{\mathsf{rpred} \mid \mathsf{rpred} \text{ is } (\Gamma, \mathcal{P})\text{-expressible}\}$ for the resulting expressible class. Importantly, we can now compare models by set inclusion: $(\Gamma, \mathcal{P})$ is (weakly) at least as expressive as $(\Gamma', \mathcal{P}')$ if $\mathsf{Expr}(\Gamma', \mathcal{P}') \subseteq \mathsf{Expr}(\Gamma, \mathcal{P})$.

Definition 2.7 (Message-Passing Neural Network (MPNN))**.** Given a graph $G = (V, E, \rho)$, a *k-layer MPNN* computes a sequence of hidden state embeddings $h_v^{(l)} \in \mathbb{R}^d$ for each node $v \in V$ and layer $l \in \{0, \ldots, k\}$. The embeddings are initialized as $h_v^{(0)} = \mathsf{INI}(\rho(v))$. For each layer $l = 0, \ldots, k-1$, the nodes update their states via a permutation-invariant aggregation over their neighborhood $N(v)$:

$$h_v^{(l+1)} = \mathsf{UPD}^{(l)}\left(h_v^{(l)}, \mathsf{AGG}^{(l)}\left(\left\{\left\{h_u^{(l)} \mid u \in N(v)\right\}\right\}\right)\right)$$

where $\mathsf{AGG}^{(l)}$ is a permutation-invariant aggregation function over multisets, and $\mathsf{UPD}^{(l)}$ is differentiable learnable functions. For k-layer MPNN m, we denote the final node embedding of v as $m(G, v) = h_v^{(k)}$.

We can easly extent the definition of MPNN to a Node Classifier:

Definition 2.8 (MPNN Node Classifier)**.** A k-layer MPNN induces a total node classifier $\mathsf{npred}_G^{\mathsf{MPNN}} : V \to \{0,1\}$ by applying a learnable readout function $\mathsf{READ} : \mathbb{R}^d \to \mathbb{R}$ to the final node embedding, followed by a thresholding indicator function:

$$\mathsf{npred}_G^{\mathsf{MPNN}}(v) = \mathbb{I}\left[\mathsf{READ}\left(h_v^{(k)}\right) > 0\right]$$

Definition 2.9 (GML)**.** *Graded Modal Logic (GML)* is a fragment of FO logic that extends standard modal logic with counting quantifiers. Its formulas are generated by boolean combinations of unary predicates and a counting modality of the form $\exists^{\geq N} y\,(E(x,y) \wedge \varphi(y))$, which expresses that there are at least N distinct neighbors y of x satisfying φ. We denote the set of GML formulas with a modal depth (nesting of modalities) of at most k as $\mathrm{GML}^{(\leq k)}$.

Definition 2.10 (Weisfeiler-Lehman (WL) Test)**.** The *1-dimensional Weisfeiler-Lehman (1-WL) test* is an iterative color refinement algorithm used to test graph isomorphism. In iteration 0, each node v is assigned an initial color based on its label $\rho(v)$. In iteration $l > 0$, the color of node v is updated injectively based on its previous color and the multiset of the colors of its neighbors.

Theorem 2.1. [**slzer2025logicalexpressivenesstemporalgnns**]
fix $k \in \mathbb{N}$ and a unary FO formula $\varphi(x)$ over a graph schema σ_G representing graphs. The following are equivalent:

- φ is realized by some k-layers MPNN node classifier.
- φ is equivalent to some $\psi(x) \in \mathsf{GML}^{(\leq k)}$

Corollary 2.1. [**grohe2022logicgraphneuralnetworks**, **xu2019powerfulgraphneuralnetworks**]
let $G = (V, E, \rho)$ be a d-labeled undirected graph and $v, u \in V$. The following are equivalent:

- forall MPNN $m : m(G, v) = m(G, u)$
- WL test gives the same color to v and u in all iterations
- forall $\psi(x) \in \mathsf{GML}^{(\leq k)} : G \models \psi(v) \Leftrightarrow G \models \psi(u)$

3 Direct and Indirect R2G are not Comperable

Definition 3.1 (Direct and Indirect R2G algorithms)**.**
fix a RDB. Direct is R2G algorithm that gives for each row in RDB a node and edge between two vertives if and only if the are joinable in RDB. Indirect is R2G algorithm that gives for each row in RDB and for each pair of column name and value (that appears in RDB) a node, and edge between a row and a column-value pair if and only if the row has that value in that column. Foramlly:
Direct R2G ($\mathcal{A}_{\mathsf{Direct}}$): The output graph is $G_{\mathsf{Direct}} = (V_{\mathsf{Direct}}, E_{\mathsf{Direct}})$, where:

- $V_{\mathsf{Direct}} = V_t$
- $E_{\mathsf{Direct}} = \{\{v_t, v_{t'}\} \mid t, t' \in Dom(RDB) \text{ and } t, t' \text{ are joinable}\}$

Indirect R2G ($\mathcal{A}_{\mathsf{Indirect}}$): Let $CV = \{(A, c) \mid \exists t \in \mathsf{Dom}(\mathsf{RDB}) \text{ s.t. } t(A) = c\}$ be the set of all attribute-value pairs present in RDB. The output graph is $G_{\mathsf{Indirect}} = (V_{\mathsf{Indirect}}, E_{\mathsf{Indirect}})$, where:

- $V_{\mathsf{Indirect}} = V_t \cup V_{cv}$, where $V_{cv} = \{v_{(A,c)} \mid (A, c) \in CV\}$ is a set of distinct vertices for each present attribute-value pair.
- $E_{\mathsf{Indirect}} = \{\{v_t, v_{(A,c)}\} \mid t \in \mathsf{Dom}(\mathsf{RDB}) \text{ and } t(A) = c\}$

Those two algorithms are used in a lot of places: Kumo [**Kumo**], and Griffin [**wang2025griffingraphcentricrelationaldata**
uses the idea from [**fey2023relationaldeeplearninggraph**] that uses the direct R2G. But on other cases like Grables [**cucumides2026grablestabularlearningindependent**] uses the indirect R2G.
Intuitively, the indirect R2G gives more information because it gives the "path" the nodes are connected. We can see this in the next lemma:

Lemma 3.1. $\mathsf{Expr}(\{\mathcal{A}_{\mathsf{Indirect}}\}, \{\mathsf{MPNN}\}) \not\subseteq \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Direct}}\}, \{\mathsf{MPNN}\})$

Sketch Proof. Considering Corolloary 2.1, it is enough to find RDB such that the WL coloring of the direct R2G is refinement of the WL coloring of the indirect R2G. Lets look on this table:

Table 1: Example of a database instance where Direct R2G fails to distinguish nodes compared to Indirect R2G.

ID	A_1	A_2
t_1	1	2
t_2	1	2
t_3	1	3

and those are the graphs after the coloring of the WL test:

Direct

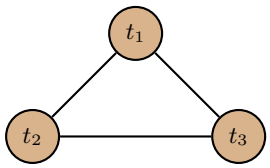

Figure 1: All tuples are assigned the same stable color (Brown) due to symmetric neighborhood structures.

Indirect

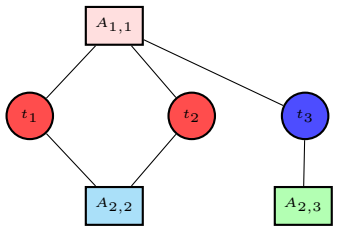

Figure 2: Distinct colors (Red, Blue) indicate that WL successfully distinguished the tuples.

■

But, suprisingly, the opposite also holds:

Lemma 3.2. $\mathsf{Expr}(\{\mathcal{A}_{\mathsf{Direct}}\}, \{\mathsf{MPNN}\}) \not\subseteq \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Indirect}}\}, \{\mathsf{MPNN}\})$

Sketch Proof. Consider the following RDB:

Table 2: Example of a database instance where Indirect R2G fails to distinguish nodes compared to Direct R2G.

ID	A_1	A_2
0	1	2
1	1	1
2	2	1
3	2	2
4	1	1
5	2	2

And the following are the graphs after the coloring of the WL test:

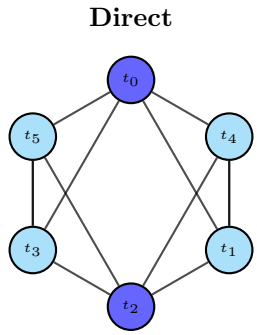

Figure 3: Distinct colors (Cyan, Blue) indicate that WL successfully distinguished the tuples.

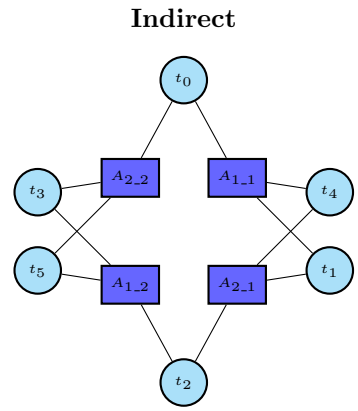

Figure 4: All tuples are assigned the same stable color (Cyan) due to symmetric neighborhood structures

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Therefore we can infer that the two algorithms are not comparable in terms of expressiveness. But can we say for what kinds of tasks one is better than the other? We will try to answer this question in the next section.

4 Separation: Where One Is Better Than the Other?

In order to find what separate between the two algorithms, we will define tow rpred that are expressible by one of the algorithms but not by the other.

Definition 4.1 ($\mathsf{rpred}_{\text{Attribute Counter}\geq N}$ and $\mathsf{rpred}_{\text{Joinable Counter}\geq N}$)**.** Let $N \in \mathbb{N}$ be a fixed threshold and RDB be a relational database instance. We define the following row predictors:

- $\mathsf{rpred}_{\text{Attribute Counter}\geq N}$: For each row t, $\mathsf{rpred}_{\text{Attribute Counter}\geq N}(\mathsf{RDB}, t) = 1$ if and only if there are at least N distinct attribute-value pairs (A, c) such that $t(A) = c$ and there exists at least one distinct row t' such that $t'(A) = c$.
- $\mathsf{rpred}_{\text{Joinable Counter}\geq N}$: For each row t, $\mathsf{rpred}_{\text{Joinable Counter}\geq N}(\mathsf{RDB}, t) = 1$ if and only if there are at least N distinct rows t' such that t and t' are joinable.

And this is our main contribution:

Theorem 4.1.

- $\mathsf{rpred}_{\text{Attribute Counter}\geq N} \notin \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Direct}}\}, \mathsf{MPNN})$
- $\mathsf{rpred}_{\text{Joinable Counter}\geq N} \notin \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Indirect}}\}, \mathsf{MPNN})$
- $\mathsf{rpred}_{\text{Attribute Counter}\geq N} \in \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Indirect}}\}, \mathsf{MPNN})$
- $\mathsf{rpred}_{\text{Joinable Counter}\geq N} \in \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Direct}}\}, \mathsf{MPNN})$

Proof. We will begin with:

- $\mathsf{rpred}_{\text{Attribute Counter}\geq N} \notin \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Direct}}\}, \mathsf{MPNN})$

Suppose, for the sake of contradiction, that $\mathsf{rpred}_{\text{Attribute Counter}\geq N} \in \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Direct}}\}, \mathsf{MPNN})$. By definition, this implies that there exists an MPNN m that exactly computes this predictor. That is, for any database instance and tuple t, m returns 1 if and only if there are at least N distinct attribute-value pairs (A, c) such that $t(A) = c$ and there exists at least one distinct row t' satisfying $t'(A) = c$.
Let $N = 2$ and consider the following database instance r:

Table 3: Counter example 1.

ID	A_1	A_2
t_1	1	2
t_2	1	2
t_3	1	3

Recall from Lemma 3.1 that the 1-WL test assigns the same stable color to t_1, t_2, and t_3 in the direct graph $\mathcal{A}_{\mathsf{Direct}}(r)$ due to their symmetric neighborhood structures. According to our initial assumption, the MPNN m must output the correct classification for each tuple:

$$m(\mathcal{A}_{\mathsf{Direct}}(r), t_1) = 1, \quad m(\mathcal{A}_{\mathsf{Direct}}(r), t_2) = 1, \quad m(\mathcal{A}_{\mathsf{Direct}}(r), t_3) = 0$$

However, by Corollary 2.1, any MPNN must output the same prediction for nodes that are indistinguishable by the 1-WL test in all iterations. Thus, m must return the same value for t_1 and t_3. This is a contradiction. Therefore, the original assertion must hold. ■

Proof.

- $\mathsf{rpred}_{\text{Joinable Counter}\geq N} \notin \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Indirect}}\}, \mathsf{MPNN})$

Suppose, for the sake of contradiction, that $\mathsf{rpred}_{\text{Joinable Counter}\geq N} \in \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Indirect}}\}, \mathsf{MPNN})$. This implies that there exists an MPNN m that exactly computes this predictor. Let $N = 4$ and consider the following database instance r:

Table 4: Example of a database instance where Indirect R2G fails to distinguish nodes compared to Direct R2G.

ID	A_1	A_2
0	1	2
1	1	1
2	2	1
3	2	2
4	1	1
5	2	2

Let us evaluate the number of distinct joinable rows for tuples t_0 and t_1. Tuple t_0 (where $A_1 = 1, A_2 = 2$) is joinable with t_1, t_3, t_4, and t_5. Thus, it has 4 distinct joinable rows, which means $\mathsf{rpred}_{\text{Joinable Counter}\geq 4}(r, t_0) = 1$. On the other hand, tuple t_1 (where $A_1 = 1, A_2 = 1$) is only joinable with t_0, t_2, and t_4. It has exactly 3 distinct joinable rows, meaning $\mathsf{rpred}_{\text{Joinable Counter}\geq 4}(r, t_1) = 0$.
Recall from Lemma 3.2 that the 1-WL test assigns the exact same stable color to all tuples (including t_0 and t_1) in the indirect graph $\mathcal{A}_{\mathsf{Indirect}}(r)$ due to their symmetric neighborhood structures. According to our initial assumption, the MPNN m must output the correct classification for each tuple, requiring:

$$m(\mathcal{A}_{\mathsf{Indirect}}(r), t_0) = 1, \quad m(\mathcal{A}_{\mathsf{Indirect}}(r), t_1) = 0$$

However, by Corollary 2.1, any MPNN must output the same prediction for nodes that are indistinguishable by the 1-WL test in all iterations. Thus, m must return the same value for t_0 and t_1, which results in a contradiction. Therefore, the original assertion must hold. ■

Proof.

- $\mathsf{rpred}_{\text{Attribute Counter}\geq N} \in \mathsf{Expr}(\{\mathcal{A}_{\text{Indirect}}\}, \mathsf{MPNN})$

Fix RDB r. To prove this statement, it suffices to construct a Graded Modal Logic (GML) formula $\varphi(x)$ such that for any vertex $v_t \in V_t$, $\varphi(v_t)$ evaluates to true if and only if row t has at least N distinct attribute-value pairs (A, c) satisfying $t(A) = c$, and there exists at least one distinct row t' such that $t'(A) = c$.
Consider the following GML formula:

$$\varphi(x) = \exists^{\geq N} y \left(E(x, y) \wedge \exists^{\geq 2} z \, E(y, z) \right)$$

We now provide a formal proof of the equivalence $\varphi(v_t) \iff \mathsf{rpred}_{\text{Attribute Counter}\geq N}(r, t) = 1$:

($\Longrightarrow$) Assume $\varphi(v_t)$ holds. By the semantics of GML, there exist at least N distinct vertices $y_1, \ldots, y_N$ adjacent to v_t. In the indirect graph construction $\mathcal{A}_{\text{Indirect}}(r)$, edges only exist between row vertices V_t and attribute-value vertices V_{cv}. Therefore, each y_i must be an attribute-value vertex of the form $v_{(A_i,c_i)}$. The edge $\{v_t, v_{(A_i,c_i)}\}$ implies that $t(A_i) = c_i$. Furthermore, the subformula $\exists^{\geq 2} z\, E(y_i, z)$ guarantees that each y_i is connected to at least two distinct vertices. Since the graph is bipartite, these neighbors must be row vertices. One is v_t, so there must exist another distinct row vertex $v_{t'} \in V_t$ ($t' \neq t$) adjacent to y_i. This implies $t'(A_i) = c_i$. Thus, there are at least N distinct attribute-value pairs shared with other rows, satisfying the condition for $\mathsf{rpred}_{\text{Attribute Counter}\geq N}(r, t) = 1$.
($\Longleftarrow$) Assume $\mathsf{rpred}_{\text{Attribute Counter}\geq N}(r, t) = 1$. By definition, there exist at least N distinct attribute-value pairs (A_i, c_i) such that $t(A_i) = c_i$, and for each, there exists a distinct row $t'_i \neq t$ such that $t'_i(A_i) = c_i$. In $\mathcal{A}_{\text{Indirect}}(\text{RDB})$, the vertex v_t is adjacent to the N distinct attribute-value vertices $v_{(A_i,c_i)}$. Let these be $y_1, \ldots, y_N$. For each y_i, there exist edges $\{y_i, v_t\}$ and $\{y_i, v_{t'_i}\}$. Since $t \neq t'_i$, y_i is adjacent to at least two distinct vertices. Therefore, the formula $\exists^{\geq N} y \left(E(v_t, y) \wedge \exists^{\geq 2} z\, E(y, z)\right)$ is satisfied.
Since $\varphi(x)$ is a valid GML formula of modal depth 2 (which is independent of the database size), Theorem 2.1 implies that there exists an MPNN node classifier that realizes $\varphi(x)$. Consequently, $\mathsf{rpred}_{\text{Attribute Counter}\geq N} \in \mathsf{Expr}(\{\mathcal{A}_{\text{Indirect}}\}, \mathsf{MPNN})$. ■

Proof.

- $\mathsf{rpred}_{\text{Joinable Counter}\geq N} \in \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Direct}}\}, \mathsf{MPNN})$

Fix RDB r. To prove this final statement, we must construct a Graded Modal Logic (GML) formula $\varphi(x)$ such that for any vertex $v_t \in V_t$ in the direct graph $\mathcal{A}_{\mathsf{Direct}}(r)$, $\varphi(v_t)$ evaluates to true if and only if there are at least N distinct rows t' that are joinable with t.
In the direct graph construction $\mathcal{A}_{\mathsf{Direct}}$, an edge exists between two vertices v_t and $v_{t'}$ if and only if the corresponding rows t and t' are joinable. Therefore, the condition that a row t is joinable with at least N distinct rows is structurally equivalent to the vertex v_t having a degree of at least N.
This property can be naturally expressed using the following GML formula:

$$\varphi(x) = \exists^{\geq N} y\, E(x, y)$$

We now establish the formal equivalence $\varphi(v_t) \Longleftrightarrow \mathsf{rpred}_{\text{Joinable Counter}\geq N}(r, t) = 1$:
($\Longrightarrow$) Assume $\varphi(v_t)$ holds. By the semantics of the counting quantifier in GML, the vertex v_t has at least N distinct neighbors $v_{t_1}, \ldots, v_{t_N}$ in $\mathcal{A}_{\mathsf{Direct}}(r)$. Since the vertices in the direct graph exclusively represent rows from the database, each neighbor v_{t_i} corresponds to a distinct row $t_i \neq t$. The existence of the edge $\{v_t, v_{t_i}\}$ directly implies that t and t_i are joinable. Thus, there exist at least N distinct rows joinable with t, satisfying the condition $\mathsf{rpred}_{\text{Joinable Counter}\geq N}(r, t) = 1$.
($\Longleftarrow$) Assume $\mathsf{rpred}_{\text{Joinable Counter}\geq N}(r, t) = 1$. By definition, there are at least N distinct rows $t'_1, \ldots, t'_N$ (all distinct from t) that are joinable with t. According to the definition of $\mathcal{A}_{\mathsf{Direct}}$, each joinable pair (t, t'_i) forms an edge $\{v_t, v_{t'_i}\}$ in the graph. Consequently, the vertex v_t is adjacent to at least N distinct vertices, which satisfies the formula $\exists^{\geq N} y\, E(v_t, y)$.
Since $\varphi(x)$ is a valid GML formula of modal depth 1 (which is constant and independent of the graph size), Theorem 2.1 guarantees the existence of an MPNN node classifier that precisely realizes this formula. This concludes the proof that $\mathsf{rpred}_{\text{Joinable Counter}\geq N} \in \mathsf{Expr}(\{\mathcal{A}_{\mathsf{Direct}}\}, \mathsf{MPNN})$. ■

5 Empirical Evaluation

6 Discussion and Conclusions